Effects of disturbance on understory plant species in Pacific Spirit Park

Updated: October 5, 2025

Background

History of Pacific Spirit Park

Pacific Spirit Regional Park has a long history. The park is located on the ancestral and unceded territory of the xʷməθkʷəy̓əm (Musqueam) nation, who occupied the land for thousands of years (land that did not look too different from our current surroundings). In the late 19th century, the area was claimed as British Columbia crown lands and xʷməθkʷəy̓əm members were forcibly relocated to a small area south of the park. After the area was logged in 1911, it was set aside as part of the University Endowment lands. Finally, in 1988, the park was officially created. This was a controversial decision that was unsupported by the xʷməθkʷəy̓əm nation at the time, though steps have since been taken to repair the relationship, including the 2008 repatriation of 22 hectares of land back to xʷməθkʷəy̓əm by the provincial government (though interestingly, the city challenged this decision in court, albeit unsuccessfully). Note that the ecological reserve area in the centre of the park (established in 1975) is accessible to xʷməθkʷəy̓əm members; several archaeological items of cultural significance have been found in that area.

Disturbances in Pacific Spirit Park

As an urban park, Pacific Spirit Park is bordered by residential areas and roads. These areas can be subject to "edge effects", in which conditions at and near the boundary of habitats differ from the interior. For example, light levels, pollution, trampling (from humans and dogs), and wind exposure are highest at edges. These are relatively stressful conditions that can affect the types of species we see and may even impact biodiversity overall.

Edge effects are most pronounced in areas of high disturbance. Just west of Pacific Spirit Park in Wesbrook Village, several large condos have been or are currently being developed, which is expected to significantly increase stressors like pollution and trampling at the forest edge. Though edge effects are typically strongest at the boundary with a highly disturbed area, trails can also create edge effects, albeit on a smaller scale.

r- vs K-selected life history traits

Plants differ in their life histories. Some strategically produce many seeds that can disperse widely, allowing their offspring to colonize newly available sites, grow fast, and reproduce early before other trees grow and shade them out (we call these "**r-selected" species**). These species tend to be relatively more tolerant of abiotic fluctuation and/or stress. Others evolve to have fewer seeds, but provide each seed with plenty of resources, allowing them to germinate in the presence of competitors. These **K-selected species** (so called because they typically exhibit logistic growth) often grow slowly but are highly shade-tolerant, increasing their competitive ability. They tend to be relatively less tolerant of stressful conditions.

How do we measure diversity?

The simplest diversity index is the **number of species** in an area (species richness), which we will call ***S***. However, if you have two hypothetical transects with the following patterns in abundance:

	Location 1 abundance	Location 2 abundance
Species A	10	5
Species B	1	5
Species C	1	5

then even though *S* = 3 for both, Location 1 primarily has individuals of just one species (10 of the 12 individuals, or 83.3%, are species A) whereas Location 2 has a more even distribution of abundance between species (each species is represented by 33.3% of individuals). Ecologists therefore have developed diversity indices that give high values when sites have lots of species and an even distribution of abundances among species. The most popular of these abundance-based diversity indices is **Shannon's diversity index (*H*)**. The formula for H is:

$$\boldsymbol{H = -\sum_{i=1}^{s} p_i * ln\,(p_i)}$$

Here, p_i is the proportion of individuals that are species *i*, and $ln(p_i)$ is simply the natural log of p_i. You already calculated components of *H* in your Excel Activity, but here's a review. The first step is to calculate $p_i * ln(p_i)$ for each species, starting with the 1st species and ending with the Sth species (where *S* = total number of species in the area). For the hypothetical Location 1 above the calculations are:

	Location 1	p_i	ln (p_i)	p_i*ln(p_i)
species A	10	0.833	-0.182	-0.152
species B	1	0.083	-2.48	-0.207
species C	1	0.083	-2.48	-0.207
sum				-0.566

The sum of $p_i * ln(p_i)$ over all the species is -0.566. Don't forget to multiply this by the negative sign in the H equation [in Excel, *(-1)] to get *H* = 0.566. If you repeat these calculations for Location 2, you would get H = 1.100, showing that once relative abundance is taken into account, site 2 is more diverse.

Research Questions

We'll be addressing four main questions in Pacific Spirit Park. First, we'll use species accumulation curves of your group's transect data to ask…

(1) did your group capture a complete sample of all species in the area?

Then, we'll use the pooled class data to investigate whether the edge of the forest affects…

(2) likelihood of pollutants?
(3) biodiversity?
(4) abundance of invasive species?

Field Methods

Materials per group

- 1 clipboard
- 1 datasheet
- 2 metre sticks
- 1 transect tape

Additional TA materials

- First aid kit
- TA clipboard with attendance sheet, background, maps, photos, permit
- Plant guide
- Compass

Protocol

1. Your TA will assign your group to a transect location. Record everyone's names, the date, and the name of your transect.
2. At a point approximately 1 m away from the trail, lay down a 20m transect using a measuring tape. At least one person should hold the handle of the tape at the trail edge and ensure the transect is staying straight and perpendicular to the trail. Avoid stepping on plants whenever possible and watch your step.
3. Once the 20m transect is laid down, group up at the **trail side of the transect (i.e., the 20 m mark)**, keeping the tape on the ground for reference. Using your 1m sticks, form a 1m x 1m square quadrat with the bottom left or right corner at the 20 metre point on the tape, and sample within that square as described in the next step.
4. Determine which species are present in the 1m x 1m quadrat and estimate their percent cover. Cover simply means the percent of the quadrat area covered by a particular species (between 0% and 100%). Estimate to the nearest 1% for values 1-20%; estimate to the nearest 5% for values >20%. Calibrate your assessment by silently estimating cover individually, then discussing as a group to land on a final value. Ignore mosses, dead plants, leaf litter, conifers, and anything taller than 2m. If one or more species in your quadrat do not appear on the plant list, ID them yourself and/or ask your TA to confirm.
   Tip: Plants may overlap with each other vertically, therefore the sum of cover values over all species may exceed 100%.
5. Once the sampling is complete, skip one meter and sample the next meter (i.e., move from 20 metres to 18 metres along the tape), moving back towards the trail. Continue on in this fashion until you have returned to the trail.
6. In total, you will sample 11 quadrats. Make note of any pollutants (dog feces, etc.). Finally, return your equipment and datasheet to your TA.

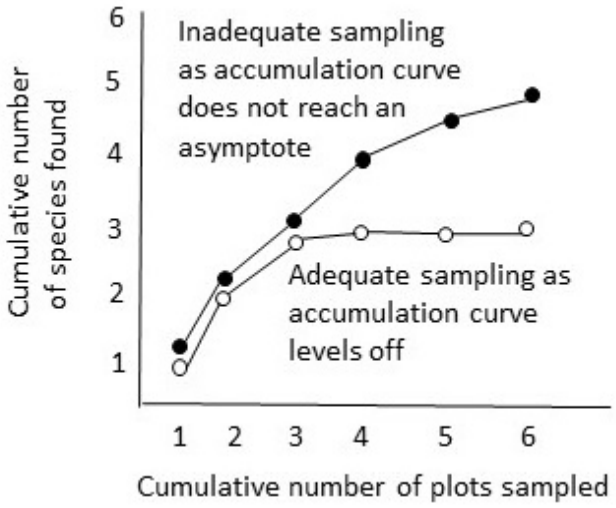
Pre-Analysis

As always, review the data sheets (species_accumulation, transect_data, diversity_data) and metadata before starting your analysis. The data do not need to be sorted for this lab.

Optional: You can use the "Freeze Panes" tool to ensure your columns headers are visible as you scroll throughout your data. Click the leftmost cell immediately below your column header, then click Window → Freeze Panes. You can always click Window → Unfreeze Panes if you freeze from the wrong cell.

Task 1: Create a species accumulation curve

In this task, you will create a species accumulation graph, displaying the cumulative number of species counted as more and more quadrats are visited. These graphs are used by ecologists to determine whether their sampling has been effective in capturing most species in the vicinity of the plot. If the sampling is effective, then after a certain number of quadrats are sampled, no (or very few) new species will be found (see righthand figure).



You will need your group's transect data. If you don't have a copy, you can download it from Canvas on the Pacific Spirit Park lab page. Once you have your group's data, you can copy the percent cover data into the sheet named "species_accumulation" (recommended so that everything is in one place) or you can work out of your group's Excel sheet.

For every quadrat sampled, you will calculate the cumulative number of unique species found. Enter these cumulative species counts in a row at the bottom of your data. Keep in mind that any % cover greater than 0 indicates that a species is present within that quadrat.

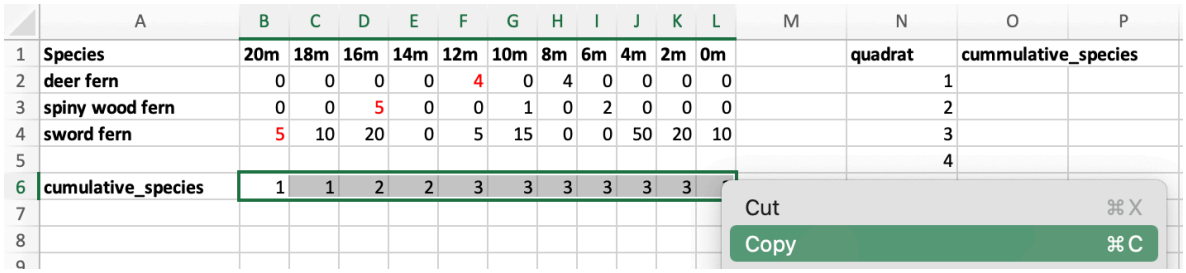
(1) In the first cell of your cumulative count row (1 quadrat visited), enter the number of species that were found at the 20m quadrat. Include the "other" (non-target) species category. In the cell to the right (2 quadrats visited), count the number of new species observed in quadrat 2 and add this to your total from quadrat 1. Repeat for quadrat 1 + quadrat 2 + quadrat 3. Continue until the final cell consists of the number of unique species found in all quadrats, as in the fake example below:

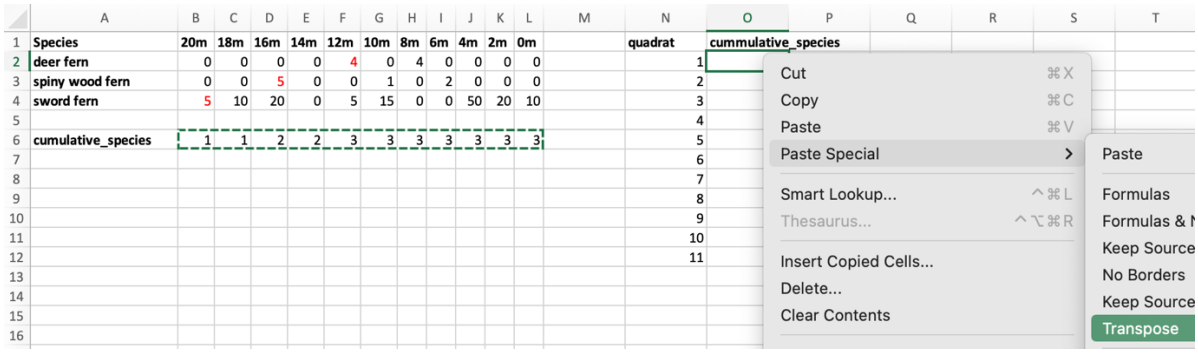
	A	B	C	D	E	F	G	H
1	**Species**	**20m**	**18m**	**16m**	**14m**	**12m**	**10m**	**8m**
2	**deer fern**	0	0	0	0	4	0	4
3	**spiny wood fern**	0	0	5	0	0	1	0
4	**sword fern**	5	10	20	0	5	15	0
5								
6	**cumulative_species**	1	1	2	2	3	3	3

Unfortunately, there is no formula that can be used to make the counting easier, so you should use any manual approach that works for you. I find it easiest to scan through each species row and highlight the first occurrence of that species within a quadrat (as in the fake example). This makes it easier to spot a unique species when I am adding up the cumulative species in each quadrat.

IMPORTANT: Make sure not to double-count species! For example, if quadrat 1 has species A, B, C, and quadrat 2 has species B, C, D, E, then the total number of unique species found in quadrat 1 + quadrat 2 is 5 (A, B, C, D, E), not 7.

(2) Next, we will plot our Y data (cumulative species) against our X data (number of quadrats sampled). To do this, most versions of Excel require that we format our data into columns. Next to the column indicating numbers of quadrats sampled (1, 2, 3, etc. up to 11), you will see an empty column for your cumulative species count. Rather than writing out your data by hand, you can rotate a copy of your row of cumulative species counts using the transpose function. After you select and copy the data you need, right-click under the count column → select "Paste Special" → click "Transpose". This will convert your horizontally-oriented data to vertically-oriented data.





(3) Create a species-accumulation curve by following the instructions from your Excel Learning Activity (chart type XY Scatter with straight lines added between the points) and formatting guidelines from the Lab Manual. It should look something like the graph on the previous page (minus the explanatory text and second accumulation curve). Name this plot Figure 1 and give it an appropriate caption. Your sample size is the number of quadrats that were surveyed.

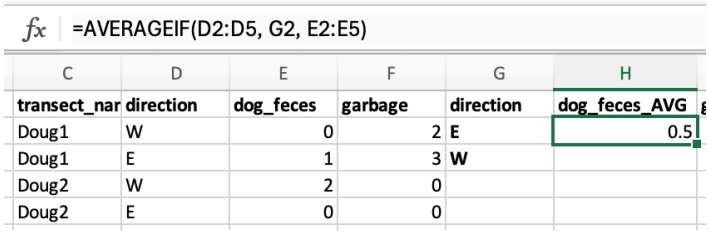
Task 2: Assess evidence of pollution

Move into the sheet named "transect_data", where you'll find columns listing out each transect, its direction (east versus west of the trail), and how much dog feces or garbage was present at a particular transect. You'll average the number of times pollution was observed on west versus east sides. Keep in mind that west-side transects are closer to the edge of the forest.

(4) We'll start by averaging the number times dog feces were recorded in west-side transects. You could do this by sorting your data and executing an AVERAGE formula on a subset of the data, but there is a faster (though slightly more challenging) way – using an AVERAGEIF formula! This is a more complex version of AVERAGE that averages based on a criterion. In our case, we want to average the number of times dog feces were observed only for data that were west of the trail ("W" in the direction column). The general syntax would be as follows:

=AVERAGEIF(full range of directions column, cell that indicates the direction you want, full range of values for a pollution column)

In the fake data below, the AVERAGEIF function averages the number of times dog feces was observed in east-side transects:



	C	D	E	F	G	H
	transect_nar	direction	dog_feces	garbage	direction	dog_feces_AVG
	Doug1	W	0	2	E	0.5
	Doug1	E	1	3	W	
	Doug2	W	2	0		
	Doug2	E	0	0		

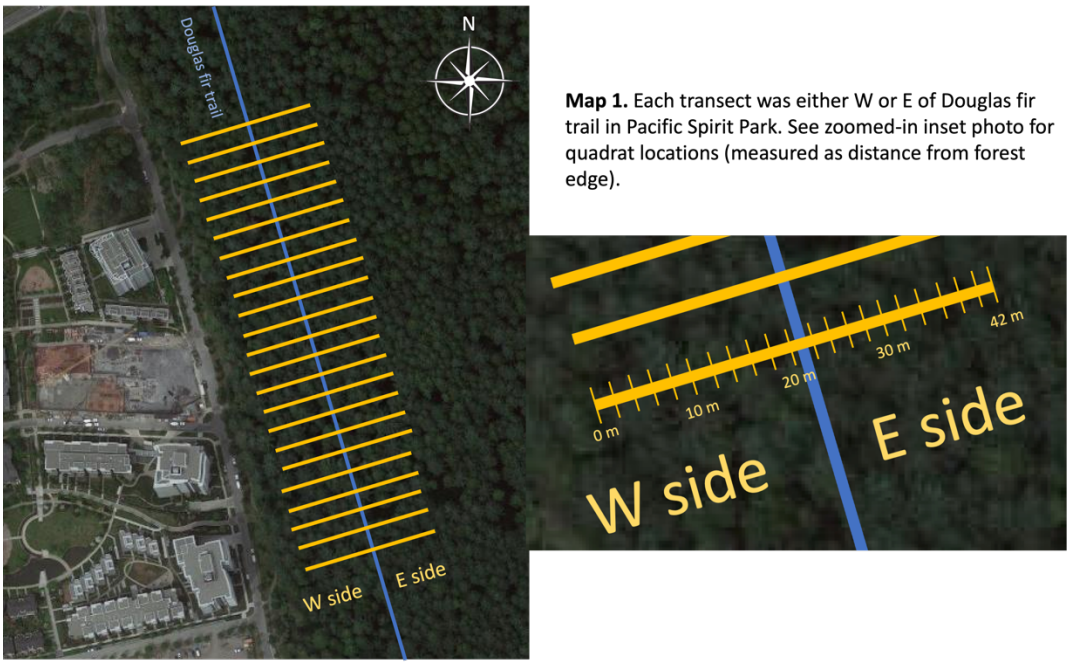
To fill in the rest of the data, you can use $ operators to fix row or column values as needed, allowing you to rapidly drag the formula to fill in most of the table on the right-hand side. Think carefully about which reference columns or rows you'd want to move as the formula is dragged, versus those you want to always stay in the same place. Return to the Excel Activity if you need a refresher on fixing and dragging.

(5) You should now have a table containing 4 values (two directions X two types of pollution). Name this table Table 1 and give it an appropriate caption. Your two sample sizes are the number of west side and east side transects, which can be determined either by selecting all the cells under "transect_name", checking the count in the bottom right corner of Excel, then dividing by two, or by executing a COUNTIF function. Report values to 2 decimal places.

Task 3: Quantifying biodiversity

Finally, move into the sheet named "diversity_data". For each quadrat location (0 metres from the forest edge, 2 metres, etc. – see Map 1 below), you will need to average percent cover of each species

across all transects. For example, all percent cover values for spiny wood fern at the 0m distance must be averaged to one single value, including zeroes. The raw data (not averaged across transects) has been provided to you on the left side of this sheet. There is also an empty table (right side of the sheet) with species columns listed in the same order, which you should use to fill in your averaged values. We'll then consider each quadrat location to be its own community and calculate biodiversity at each one.



Map 1. Each transect was either W or E of Douglas fir trail in Pacific Spirit Park. See zoomed-in inset photo for quadrat locations (measured as distance from forest edge).

Optional note (for clarification, not necessary for analysis): quadrat locations were renamed based on distance to forest edge. For example, a distance of 42 m from the edge would have originally been recorded as 0m on the east side of the trail, 22 m was originally 20 metres east (i.e., the point closest to the trail for east-side transects), and 20 m was originally 20 metres west (i.e., the point closest to the trail for west-side transects).

(6) First, we will average the percent cover for each species (spiny wood fern, sword fern, etc.) at each quadrat location (0 m, 2 m, etc.) across all transects. To do this, we can use another AVERAGEIF function. The syntax is the same as in Task 2, but we'll apply it to different data:

=AVERAGEIF(full range of distance column in the raw data, cell that indicates the distance you want, full range of values for a species % cover in the raw data)

This function is powerful because it allows us to write just one formula, fix some key values with $ operators, and then drag to fill out the entire table in a few minutes. If you tried to use a normal AVERAGE function, it would take ages!

In the fake data below, we've used the AVERAGEIF function to calculate the average percent cover of salal at 0 m across three transects. Consider how you could modify the formula so that it could be dragged right and down to automatically fill in the rest of the right-hand table.

H2 =AVERAGEIF(B2:B7, G2, C2:C7)

	A	B	C	D	E	F	G	H	I	J
1	transect_name	distance_m	salal	starflower	foamflower		distance_m	salal_avg	starflower_avg	foamflower_avg
2	Doug1	0	1.5	0	1.5		0	8.833333		
3	Doug1	2	15	0	3		2			
4	Doug2	0	25	0	0					
5	Doug2	2	8	0	0					
6	Doug3	0	0	0	2					
7	Doug3	2	0	1	1					

Once you've calculated the average % cover for every combination of species and quadrat location, we'll fill in some new data columns towards the right. Use your averaged data for all the following calculations, not the raw data on the left.

(7) First, use the SUM function to add up the total % cover across all species at the 0m distance (column "total_cover"). This gives us total abundance of the 0m community, which will allow us to calculate relative abundances in the following steps. Once you have the summed % cover for 0m, drag to fill in the summed cover values for all the other quadrat locations.

(8) Next, we'll measure how biodiversity changes with increasing distance from the forest edge. Calculating the Shannon index more than one or two times in Excel is prohibitively onerous, so I have calculated all the values for you except one (lucky you!). Using what you learned in class and in the lab background, calculate the Shannon index for the community at 42 m. To calculate a natural logarithm, you can use the function LN().

Hint #1: You will need to calculate relative abundance for each species – think about how you can use the data from the previous step to do this.

Hint #2: It is possible to do this calculation in one step using brackets, but you may find it easier to calculate across multiple steps – think about where you might be able to add new rows or columns to write out each step.

Hint #3: For species that have a percent cover of 0 at 42 m, you will need to manually enter NA or exclude them from your calculations, otherwise you will get an error (cannot take the ln of 0!).

(9) Insert an XY scatter plot to visualize how biodiversity (Shannon) changes with increasing distance from the forest edge (distance_m). This plot will be named Figure 2. You should add a trendline and display the R-squared value (return to your Excel Activity if you forget how to do this). Note also that the Y-axis may automatically range from 0 to 2 or 3, but most of our values will be well above 0. If you feel the trend is being obscured, you could optionally adjust the y-axis range by double-clicking or right-clicking on the y-axis and setting a more appropriate minimum bound under Axis Options.

Format it nicely and add a caption. You will need to report the Shannon index for 42 m in the caption (round to 2 decimal places). Your sample size is the total number of quadrat locations and can be determined using similar approaches to Task 2 step 5.

(10) Finally, we'll consider how the relative abundance of invasive species changes with increasing distance from the forest edge. In the invasive_proportion column, calculate the summed cover of our invasive species only (English holly, Himalayan blackberry, English laurel, and wall lettuce) divided by total cover of all plants.

Plot these data as in the previous step to create Figure 3, making sure to include a trendline, R-squared, and caption.

Instructions for lab report on next page

Lab Report [35 MARKS TOTAL] - Due Friday October 11 at 5:00 pm (buffer to 11:59 pm)
Page 1 and 2 of your report consists of only Figure 1, Table 1, Figure 2, and Figure 3. ***The last page is only for answering the questions and must not exceed one page****. See lab manual for details on formatting and general guidelines.*

Please also submit your Excel spreadsheet.

Figures and Tables [15 marks]

- Species accumulation curve for your group's transect (Figure 1) [4 marks]
- Average count of dog feces and garbage in East versus West transects (Table 1) [3 marks]
- Shannon index against distance from forest edge (Figure 2) [4 marks]
- Total invasive species cover against distance from forest edge (Figure 3) [4 marks]

Questions [17 MARKS]

Read the background information to guide your responses. If you accessed any sources outside of course materials, list those sources at the end of your report (will not contribute to page count; formatting not necessary) or in a comment on your Canvas submission. Refer to the syllabus for policies on appropriate use of generative AI tools and see below for a special note on question 5b.

Part I: Effectiveness of sampling

1. Compare your Figure 1 to the plot on page 4 of this lab. Do you believe the species at your group's transect were adequately sampled? Justify your answer, to the best of your ability. [3 marks]

Part II: Edge effects

2. Summarize the results in Table 1. Were there differences between west transects (closer to the edge of the forest) and east transects (farther from the edge of the forest)? Did the findings match what you'd have predicted, based on what you know about the park and your personal observations during field work? [2 marks]

3. Based solely on the background information and methodology (i.e., *not* the results), state and justify two specific, contrasting, and mutually exclusive hypotheses for how you'd expect biodiversity to change (or not) with increasing distance from the edge of the forest. Then, focusing on the trendline and R-squared, summarize the main takeaway from Figure 2. Is there a hypothesis that is clearly better supported by your results? Alternatively, is there partial support for both hypotheses, or no support for either hypothesis? [5 marks]
   Hint: if you need help interpreting the R-squared, return to the Excel Activity or post on Piazza. Remember that R-squared tells us the percentage of variation in Y can be explained by X. Another more quantitative approach is to calculate a Pearson's r correlation coefficient by taking the square route of the R-squared, assigning it a positive or negative sign based on the direction of your trendline, and calculating a P-value. The P-value helps you determine whether the relationship is

significant; i.e., is the correlation truly different from a correlation of zero, or could we see a value at least as extreme as this one due to random chance? Enter your correlation coefficient and sample size into the text boxes here. Interpret the two-tailed P-value only. If the P-value is <0.05, the trend is significant and likely did not occur due to chance. If the p-value is >0.05, it is not significant, meaning that any apparent correlation you see in the data is statistically indistinguishable from no correlation at all.

4. Identify two life history traits you'd expect an invasive species to have. Based on these traits, would you expect to see a higher abundance of invasive species at the edge of the forest or deep inside the forest? Then, focusing on the trendline and R-squared, summarize the main takeaway from Figure 3. Are these results surprising, or do they match what you'd expect? [4 marks]

Part III: Species-specific patterns

5. Choose one of the native plant species from our study, then consult some reliable sources* to learn about its life history traits and cultural significance. [3 marks total]

    a. Is your chosen species broadly r-selected or K-selected? Justify your answer based on two of its traits (think of some of the traits we've discussed in class and in the lab background to get an idea of what to look for). [2 marks]

    b. Briefly indicate one way in which this plant is important to local first nations (such as Musqueam, xʷməθkʷəy̓əm). Its importance can be medicinal, used in food, cultural, spiritual, etc. Please keep your response respectful and use present-tense language (unless you are sure the importance/usage is solely historical) and do not use AI, even for brainstorming**. [1 mark]

* Reliable resources include (but are not limited to):

- "Ecological Indicator" information on UBC's E-Flora website: http://ibis.geog.ubc.ca/biodiversity/eflora/E-FloraTreesofBritishColumbia.html
- Fire Effects Information System: https://www.feis-crs.org/feis/
- Plant guides, such as:
    - Food Plants of Coastal First Peoples (Turner)
    - Plants of Coastal British Columbia (Pojar and Mackinnon)
      (Rachel has copies of both, if you want to take a look)
- Sierra Club of BC: https://sierraclub.bc.ca/ecomap/
- First Nations Traditional Food Fact Sheet: https://www.fnha.ca/Documents/Traditional_Food_Fact_Sheets.pdf

** It's important to note that many Indigenous people feel the use of generative AI to report on Indigenous knowledge is appropriative and harmful, since programs like chatGPT may be accessing private sources where permission to share knowledge was not given. Please keep this in mind and see here for an example of this perspective: https://techpolicy.press/an-indigenous-perspective-on-generative-ai/